

Sonifying the Heart: Inner Emotion Detection from ECG through Music

Music Workshop Sprint 3: Individual Contribution

Sathwik Reddy 2024121002

1 My Role

I built the fourth stage of our pipeline, **Emotion**, which reads the emotion of the generated music. I also evaluated it.

2 What I Did

- **Our emotion model.**
 - It places each melody on Russell’s valence–arousal plane, using three musical cues: rhythmic irregularity, pitch spread and consonance.
 - Tempo and loudness are left out on purpose. Tempo differs more between people than between rhythms, and loudness is re-scaled within each clip.
- **Essentia.**
 - Four pretrained mood classifiers and a valence/arousal model trained on DEAM, run on both the melody and the full arrangement.
 - Found and fixed a bug: the classifiers list their classes in different orders (**happy**, **non_happy** but **non_sad**, **sad**).
- **music2emo.** Set up its separate environment and ran it on all 14 melodies.
- **Comparison.** For every recording, checked whether each method scores the AFib piece higher or lower than the normal one.
- **Validation plan.**
 - ECG datasets with self-reported emotion (WESAD, DREAMER, AMIGOS), compared against plain heart-rate variability.
 - The ethics of using this with specific groups, such as autistic people.

3 Key Result

Our model reads all 7 normal pieces as calm, and 6 of the 7 AFib pieces as tense.

All three methods hear a difference in every recording, but no two agree on both valence and arousal:

- Essentia hears AFib as more pleasant and more energetic.
- music2emo hears it as less of both.

Tempo probably explains part of this, since every AFib clip is also the faster heart.

4 Next Steps

- Run the validation on WESAD first.
- Design the listening study.